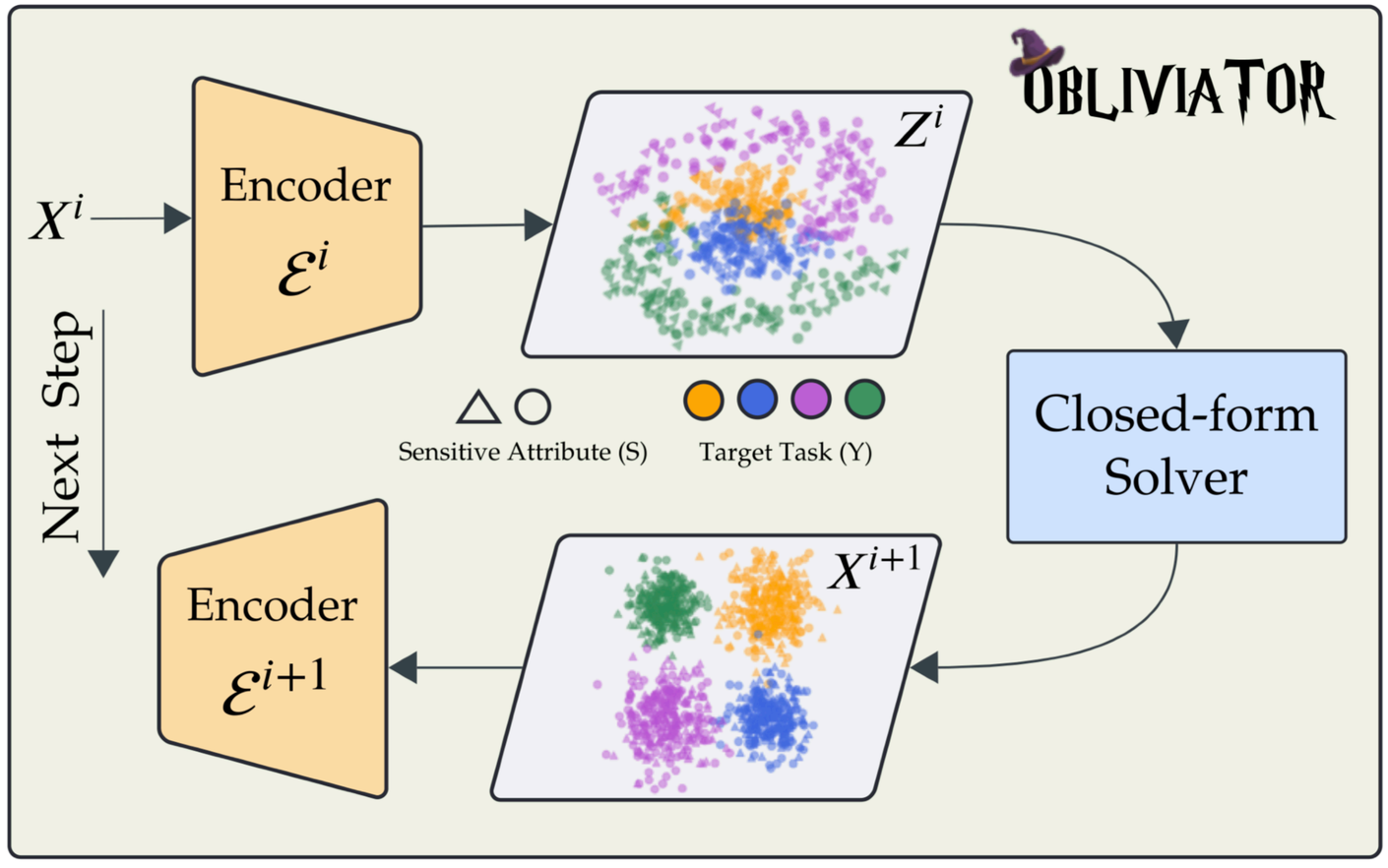
How Do We Solve This Optimization ?



Step Two

Separate Useful Information from Unwanted Concepts:

$$\sup_{\{g_a \in \mathcal{G}_a\}} \sup_{f \in \mathcal{F}} \mathbf{Cov}^2(f(Z_\theta^i), g_y(Y)) + \mathbf{Cov}^2(f(Z_\theta^i), g_{x^i}(X^i)) + \mathbf{Cov}^2(f(Z_\theta^i), g_x(X))$$

$$s.t \quad \sup_{g_s \in \mathcal{G}_s} \mathbf{Cov}(f(Z_\theta^i), g_s(S)) = 0 \quad \|f\|_{\mathcal{F}} = \|g_a\|_{\mathcal{G}_a} = 1$$

Single Stage Vs. Iterative Erasure